

Core 1 Concept Rescue

For Brandon | A+ 220-1201 | Reviewed September 21, 2026

See the idea. Say it in your own words. Solve the scenario.

This guide targets your core topics plus SCSI and connection throughput. Each chapter has a visual explanation, a comparison page, and three original multiple-choice questions with explanations. Use it to build understanding rather than memorize answer letters.

Topic	Pages	Related objectives
1. Cloud types	2-4	4.2
2. Server types	5-7	2.1, 2.3, 2.4
3. IP addresses	8-10	2.6
4. Cable categories	11-13	3.2, 2.8
5. Routers and switches	14-16	2.5, 2.4, 5.5
6. Wi-Fi standards	17-19	2.2, 5.5
7. RAID	20-22	3.4, 5.2
8. Connection types	23-25	3.2, 2.7
9. PCI cards	26-28	3.5, 3.4
10. SCSI and SAS	29-31	3.4, 3.5; legacy context labeled
11. Connection throughput	32-34	3.2, 3.4, 3.5, 5.5

TONIGHT: one useful pass

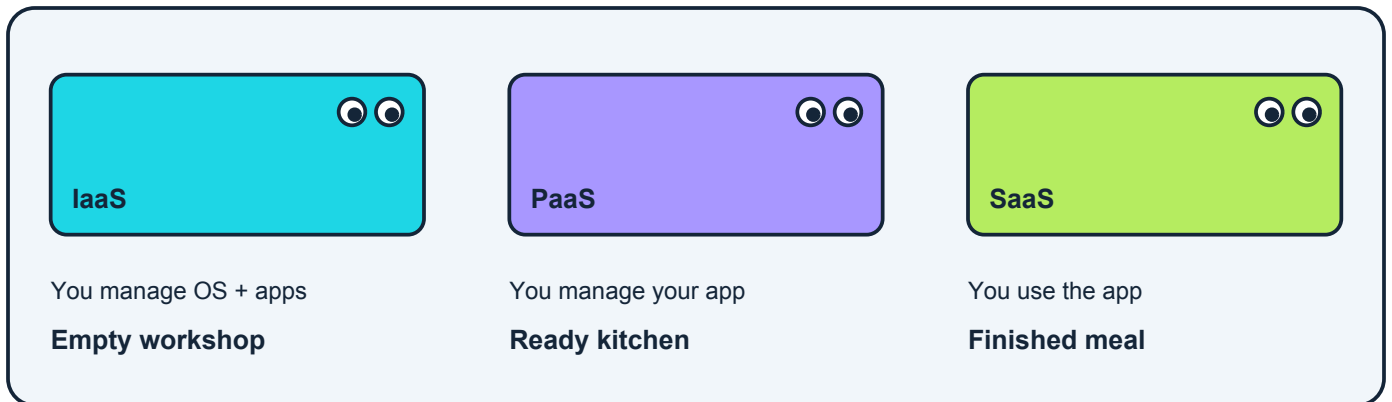
Start with IP addresses, servers, and routers because they connect. Then tackle cloud and RAID. Finish cables, Wi-Fi, connectors, and PCIe. Try each quiz before reading its answers. Watch the linked lesson only where the idea is still fuzzy. Leave time to rest.

Scope: public CompTIA 220-1201 objectives, with primary technical references. These are original practice examples, not official exam questions or a promise of exam coverage. Distances use approximate feet/inches; standard names keep their usual labels.

Cloud types: who uses it, who manages it?

Related Core 1 objectives: 4.2

Visual: Three bright cloud-service characters show increasing provider responsibility: IaaS, PaaS, and SaaS. Compare each character with the workshop, kitchen, and meal analogy.



Cloud computing gives you computing resources over a network when you need them. Think of renting useful tools instead of buying every machine. A cloud can supply storage, processing power, software, or a place to run your code. A server sitting somewhere else is not automatically a complete cloud service.

Two separate questions. A deployment model asks WHO gets to use the cloud: public, private, community, or hybrid. A service model asks WHAT the provider manages: IaaS, PaaS, or SaaS. One service can be both public cloud and SaaS.

IaaS: Infrastructure as a Service. Rent the workshop. The provider supplies physical hardware and virtualization. You typically manage the guest operating system, installed applications, and data. Example: rent a virtual machine and install your own server software.

PaaS: Platform as a Service. Rent the ready kitchen. The provider manages the infrastructure, operating system, and application runtime. You deploy your code and manage its data and settings. A runtime is the software environment that runs your program.

SaaS: Software as a Service. Receive the finished meal. Use the provider's complete application, such as hosted email. You still manage your account, permissions, and how you use or share data. SaaS does not erase your security responsibilities.

WATCH ON YOUTUBE - Professor Messer

[Cloud types - A+ 220-1201](https://www.youtube.com/watch?v=KZxAY5ssUxc)

<https://www.youtube.com/watch?v=KZxAY5ssUxc>

The details that separate the answers

Deployment	Who uses it?	Scenario clue
Public	Customers use a provider's offered services.	A business rents cloud resources available to many customers.
Private	One organization has exclusive use.	A company's cloud serves only its departments; hosting can be on or off site.
Community	Organizations with shared requirements.	Several hospitals share a cloud designed for their common rules.
Hybrid	Distinct clouds work together.	A private cloud connects to a public cloud for extra capacity.

Elasticity versus scalability. Scalability is the ability to handle more work by adding capacity. Elasticity adjusts capacity up AND down as demand changes, often automatically. Picture extra checkout lanes opening for a rush and closing afterward.

Metered use, ingress and egress. Metering measures use, such as processing hours or storage. Ingress is data entering a service; egress is data leaving it. Charges depend on the provider and plan. Do not assume every upload or download is free.

Shared resources and multitenancy. Multiple customers can share underlying equipment while remaining logically separated. A tenant is a customer or organization using the service. Dedicated resources are reserved for one customer; that does not automatically make the whole service a private cloud.

Availability and synchronization. Availability means the service is ready when needed. Redundant systems can improve it. File synchronization keeps copies in step across devices. A deletion may sync too, so synchronization alone is not a safe replacement for backup.

Virtualization connection. A virtual machine (VM) is a software-defined computer. A hypervisor manages VMs. Type 1 runs directly on hardware; Type 2 runs on a host operating system. Virtualization can help build clouds, but a VM on your laptop is not automatically a cloud.

Can you explain your choice?

Original Core 1-style practice | Select ONE answer per question

EXAM TRAP

Public describes who can obtain the service, not who can read your files. Private does not have to mean inside your building. A brand name alone does not tell you whether a product is IaaS, PaaS, or SaaS.

1. A developer uploads code. The provider patches the operating system and runtime. Which service is this?

- A. IaaS
- B. PaaS
- C. SaaS
- D. RAID

2. A private cloud adds capacity from a connected public cloud during a sales rush. Which deployment fits?

- A. Private only
- B. Community
- C. Hybrid
- D. Local storage

3. Employees use a provider's finished online email application. Which service model fits?

- A. SaaS
- B. PaaS
- C. IaaS
- D. Type 1 hypervisor

Answers - check only after you try

1. B. PaaS supplies the managed platform while the customer supplies the application. IaaS leaves the guest OS to the customer; SaaS supplies a finished application.

2. C. Hybrid combines distinct clouds. The capacity rising and falling is also an example of elasticity; that is a characteristic, not the deployment model.

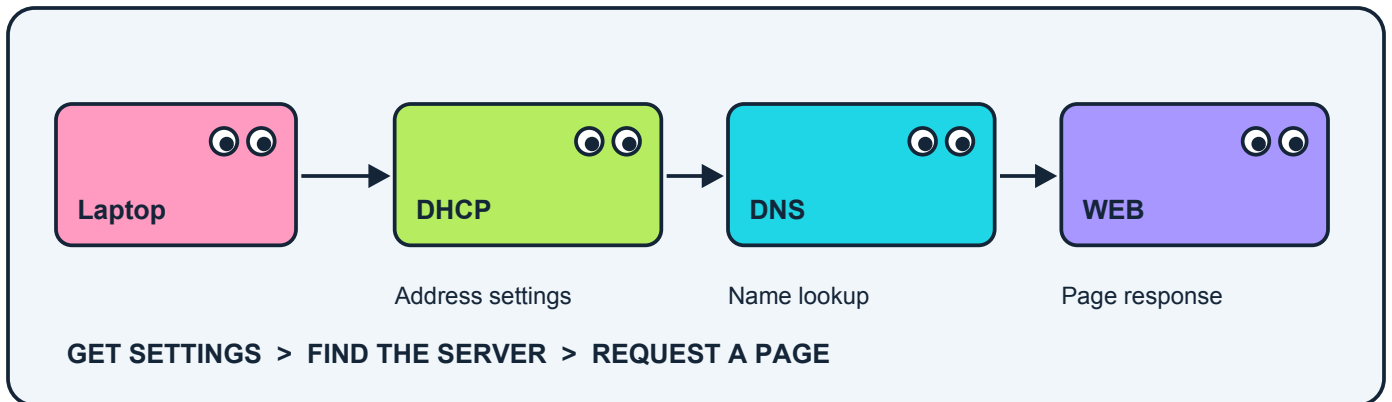
3. A. They use a complete application, so this is SaaS. They are not installing an operating system or deploying application code.

Say it back: explain the right answer without naming its letter. Then explain why one other option does not fit. If you can do that, you are learning the concept.

Server types: the helpers on your network

Related Core 1 objectives: 2.1, 2.3, 2.4

Visual: A lime DHCP clerk hands out address badges. A cyan DNS librarian looks up names. A purple web server hands a page to a smiling laptop.



A server is a system or program that provides a service. A client asks for that service. The word describes a job, not just a large metal box. One physical computer can run several server roles; one role can also run on several computers for reliability. Tower, rack, and blade describe physical server designs, not their network jobs.

DNS: Domain Name System. The name librarian. It finds records for names. An A record gives an IPv4 address; an AAAA record gives an IPv6 address. CNAME points a name to another name. MX identifies mail servers. TXT holds text used by several services and policies.

DHCP: Dynamic Host Configuration Protocol. The address clerk. It leases an IP address and can supply a subnet mask, gateway, and DNS servers. A lease lasts for a set time. A scope defines an address pool and settings. A reservation repeatedly offers a particular client the same address.

The first conversation. A common initial IPv4 DHCP exchange is DORA: Discover, Offer, Request, Acknowledge. The client looks for help, a server offers settings, the client requests them, and the server confirms the lease.

DNS versus DHCP. If you already have valid addressing but a name will not resolve, investigate DNS. If automatic addressing fails, investigate the link and DHCP. DNS does not normally hand out the client's IP address.

WATCH ON YOUTUBE - Professor Messer

[Server types - A+ 220-1201](https://www.youtube.com/watch?v=Z-9mkqi2ELI)

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The details that separate the answers

Server / service	Job and exam clue	Useful default
DNS	Find records for names.	TCP/UDP 53
DHCP (IPv4)	Lease addresses and settings.	UDP 67 server / 68 client
Web	Answer browser requests.	HTTP 80 / HTTPS 443
File share	Central access to shared files.	SMB TCP 445
Mail	Send or retrieve email.	SMTP 25; POP3 110; IMAP 143
Print	Queue and manage print jobs.	Protocol varies
AAA	Verify identity, decide access, log activity.	Protocol varies
Database	Store/query structured records.	Product-specific
NTP	Synchronize clocks.	UDP 123
Syslog	Collect event messages.	Often UDP 514

AAA: three different jobs. Authentication: Who are you? Authorization: What may you do? Accounting: What did you do? A successful sign-in does not grant permission to every shared folder.

Mail and port meaning. SMTP means Simple Mail Transfer Protocol and sends mail; port 25 is commonly server-to-server. POP3 means Post Office Protocol version 3; IMAP means Internet Message Access Protocol. Both retrieve mail; IMAP supports server-based mailbox synchronization. Secure variants and submission ports differ.

Appliances you may confuse with servers. A load balancer distributes requests across healthy back-end servers. A proxy acts on behalf of a client or server. A spam gateway filters unwanted email. Unified Threat Management (UTM) combines several security functions in one system.

Other environments. Internet of Things (IoT) devices include connected sensors and appliances. Supervisory Control and Data Acquisition (SCADA) monitors and controls industrial processes. These devices may need careful segmentation and vendor-supported maintenance.

How to choose a role. Start with the job in the question. Address lease = DHCP. Name lookup = DNS. Shared documents = file server. Many pending jobs = print queue. Matching clock time = NTP. Central event records = Syslog.

Can you explain your choice?

Original Core 1-style practice | Select ONE answer per question

EXAM TRAP

A port is a logical service door, not the physical jack on the computer. TCP and UDP are transport protocols: Transmission Control Protocol and User Datagram Protocol. TCP provides ordered, reliable delivery; UDP avoids that built-in connection/retransmission mechanism.

1. A PC reaches a test service by IP address, but its hostname does not resolve. Which service should be checked first?

- A. DHCP
- B. Print
- C. DNS
- D. NTP

2. Several printers must receive a predictable address through DHCP. What should you configure?

- A. A reservation for each printer
- B. A new MX record
- C. A bigger print queue
- D. Random static addresses inside the pool

3. An administrator needs network devices to send event messages to one central collector. Which service fits?

- A. IMAP
- B. Syslog
- C. DHCP
- D. SMTP

Answers - check only after you try

1. C. The clue points to name resolution. Working IP connectivity makes a basic address-leasing problem less likely, although a real investigation still checks the actual error.

2. A. A reservation maps a client identifier, often its MAC address, to a repeatable DHCP address. Avoid accidental overlap with manually assigned addresses.

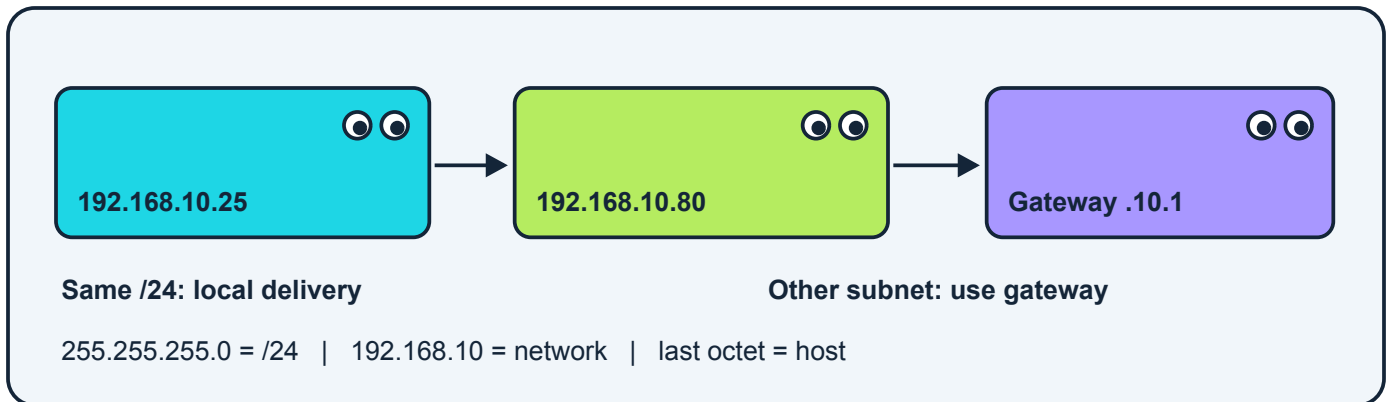
3. B. Syslog collects event messages. NTP helps align timestamps, but it is not the event collector.

Say it back: explain the right answer without naming its letter. Then explain why one other option does not fit. If you can do that, you are learning the concept.

IP addresses: find the neighborhood first

Related Core 1 objectives: 2.6

Visual: Two bright address blocks share the 192.168.10.0/24 neighborhood. A purple gateway provides the next hop toward other networks.



IP means Internet Protocol. An IP address identifies a network interface for IP communication. Think of the address as a delivery location. The subnet mask tells the computer which part identifies the neighborhood. The default gateway is the next stop for destinations outside that neighborhood.

IPv4: four numbers. Internet Protocol version 4 uses 32 bits. We write four octets, such as 192.168.10.25. An octet has 8 bits and a decimal value from 0 through 255. An address containing 300 is not a valid IPv4 address.

Subnet mask and /24. The mask 255.255.255.0 means the first 24 bits identify the network. Slash notation /24 is called a prefix length. In this example, 192.168.10 is the network part and the last octet identifies a host. This shortcut works here because the prefix is /24; other masks change the boundary.

Local or remote?. 192.168.10.25/24 sends to 192.168.10.80 directly across the local network. To reach 192.168.20.80, it normally sends the packet to a router. The gateway needs an address reachable on the local subnet, such as 192.168.10.1.

MAC is a different address. A Media Access Control (MAC) address identifies an interface at the local-link level. A switch uses MAC information to forward Ethernet frames. A router uses IP information to choose a route. ARP, Address Resolution Protocol, finds an IPv4 neighbor's MAC address.

WATCH ON YOUTUBE - Professor Messer

[IP addresses - A+ 220-1201](https://www.youtube.com/watch?v=yubEz-ZEVwY)

<https://www.youtube.com/watch?v=yubEz-ZEVwY>

The details that separate the answers

Address type	Range or example	Meaning
Private IPv4	10.0.0.0 - 10.255.255.255	Private range: 10.0.0.0/8
Private IPv4	172.16.0.0 - 172.31.255.255	Only this part of 172.x is private.
Private IPv4	192.168.0.0 - 192.168.255.255	Private range: 192.168.0.0/16
IPv4 link-local	169.254.0.0/16	APIPA commonly appears after DHCP fails.
IPv4 loopback	127.0.0.0/8; usually 127.0.0.1	Refers back to the local host.
IPv6	128 bits; e.g. 2001:db8::25	Hexadecimal groups; example prefix reserved for documentation.
IPv6 link-local	fe80::/10	Used on the local link; normal in IPv6.
IPv6 loopback	::1	The local host.

Public, private, static, dynamic. Public/private describes address use and scope. Static/dynamic describes assignment. They are separate pairs: a private address can be static or leased. Not every non-private address is usable public space; loopback and other special ranges exist.

APIPA: Automatic Private IP Addressing. Windows may self-assign a 169.254 address when it cannot obtain an IPv4 DHCP lease. This supports local-link communication, not normal routed internet access. Check the cable or Wi-Fi association, DHCP reachability, and available leases. It does not prove the DHCP server itself is broken.

A complete /24 example. Host: 192.168.10.25. Mask: 255.255.255.0. Network: 192.168.10.0. Broadcast: 192.168.10.255. Typical usable hosts: .1 through .254, or 254 addresses. Do not assign the network or broadcast address to a normal host in this subnet.

IPv6 reading rules. There are eight groups of up to four hexadecimal digits. Leading zeros within a group can disappear. One consecutive run of zero groups can become ::, only once in an address. A common LAN prefix is /64; not every IPv6 network uses that prefix. IPv6 does not use broadcast.

NAT: Network Address Translation. A home router commonly translates private IPv4 traffic to a public address; port translation lets several clients share it. NAT is not DNS, and it is not a substitute for firewall policy.

Can you explain your choice?

Original Core 1-style practice | Select ONE answer per question

EXAM TRAP

172.15.1.5 is not in the 172.16/12 private range; 172.20.1.5 is. A .1 gateway is a common choice, not a rule. Having Wi-Fi bars proves neither a valid IP configuration nor working internet access.

1. Which address belongs to an RFC 1918 private range?

- A. 172.15.4.8
- B. 172.20.4.8
- C. 169.254.4.8
- D. 127.0.0.1

2. A host is 192.168.4.20/24. Which destination is local?

- A. 192.168.5.20
- B. 192.168.40.1
- C. 192.168.4.90
- D. 192.169.4.20

3. After a router change, a DHCP-configured PC has 169.254.8.9. What should you investigate?

- A. The monitor resolution
- B. Only DNS records
- C. A failed RAID mirror
- D. Network/DHCP reachability

Answers - check only after you try

1. B. The private 172 range runs from second octet 16 through 31. The 169.254 and 127 ranges have different special purposes.

2. C. With /24, the first three octets must match for this local subnet. The other destinations require routing.

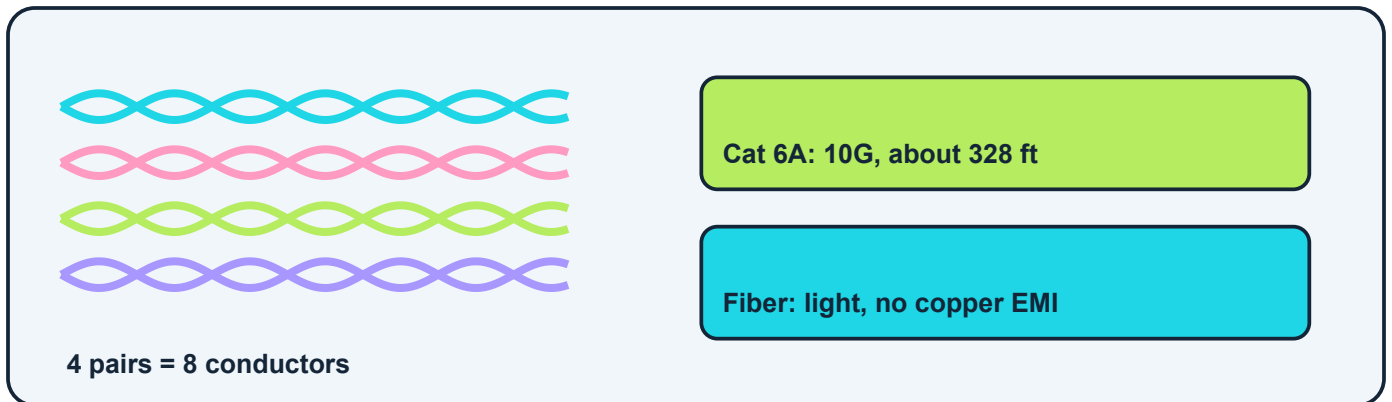
3. D. The link-local address suggests the client did not obtain a DHCP lease. Verify the path and service before deciding which component failed.

Say it back: explain the right answer without naming its letter. Then explain why one other option does not fit. If you can do that, you are learning the concept.

Cable categories: choose the whole path

Related Core 1 objectives: 3.2, 2.8

Visual: Four twisted copper pairs race along a neon track. A shield blocks noise. A glass-fiber runner sends light past an electrical storm.



A cable is a path for signals. Copper carries electrical signals; fiber carries light. A cable category describes tested performance requirements. It does not magically set the speed of your internet plan, your switch, or your network card.

Twisted pair. Ethernet copper cable commonly contains four pairs, or eight conductors. Twists help reduce unwanted noise. Crosstalk is interference between pairs or cables. EMI means electromagnetic interference, such as noise from nearby electrical equipment.

UTP versus STP. Unshielded Twisted Pair (UTP) has no overall shielding requirement implied by that name. Shielded Twisted Pair (STP) adds shielding. Shielded systems need the proper installation and bonding/grounding. Shielding and a fire rating are different properties.

Speed words. Mbps means megabits per second; Gbps means gigabits per second. Eight bits make one byte. A 1 Gbps link is not a 1 GB-per-second file transfer. Real transfers lose some capacity to protocol overhead and other limits.

A complete channel. The common maximum copper Ethernet channel is about 328 ft, including patch cords. A typical structured layout allows about 295 ft of permanent cable plus about 33 ft total of patch cords. The entire channel must meet the needed specification.

WATCH ON YOUTUBE - Professor Messer

[Cable categories - A+ 220-1201](https://www.youtube.com/watch?v=29X5Ho3m2KU)

<https://www.youtube.com/watch?v=29X5Ho3m2KU>

The details that separate the answers

Category	Common study association	Distance / caution
Cat 5 (legacy)	100 Mbps; compliant Cat 5 can also support 1000BASE-T.	Do not treat 100 Mbps as an absolute technical ceiling.
Cat 5e	1 Gbps over a compliant channel.	Up to about 328 ft.
Cat 6	1 Gbps full channel; 10 Gbps on shorter suitable runs.	10G often summarized as up to about 180 ft; test crosstalk.
Cat 6A	10 Gbps over the full channel.	Up to about 328 ft. A = augmented.
Cat 8 (extra context)	25/40 Gbps for short data-center copper links.	About 98 ft; lower priority than 5e/6/6A here.

Choose by requirement. For a new 10 Gbps copper run around 300 ft, Cat 6A is the clear choice among Cat 5e, Cat 6, and Cat 6A. For an ordinary 1 Gbps run under 328 ft, Cat 5e can meet the requirement. Faster cabling does not upgrade a 100 Mbps port.

T568A and T568B. These define conductor order at eight-position connectors. Same scheme at both ends makes a straight-through cable. A on one end and B on the other makes a traditional crossover. Many modern ports use auto-MDI-X to adapt automatically. Match the installation standard and test the wire map.

Fire and outdoor ratings. Plenum cable has a rating for applicable air-handling spaces, with flame/smoke requirements. Riser cable has a different rating for vertical pathways. Direct-burial cable is designed for installation in the ground. These labels are not speed categories or automatic substitutes for one another.

Fiber and coax. Single-mode fiber uses a small core and supports long-distance optical links. Multimode uses a larger core and is common for shorter building/data-center links. Distance depends on fiber type and transceivers; there is no universal fiber limit. Coaxial cable has a center conductor and surrounding shield; cable broadband commonly uses an F-type connector.

Use the right tool. Crimper: attach compatible modular plugs. Punchdown tool: seat wires in terminals on a jack or patch panel. Toner/probe: trace a cable. Cable tester: check faults and wire map. A basic continuity test does not certify a cable for 10 Gbps.

Can you explain your choice?

Original Core 1-style practice | Select ONE answer per question

EXAM TRAP

Cat 6 is not automatically 10 Gbps at 328 ft. Cat 6A is the full-length 10G study answer. Never look into a fiber end; invisible optical energy may be present. Keep fiber scraps and sharp edges away from eyes and skin.

1. A new 300 ft copper channel must support 10GBASE-T. Which category is the best listed choice?

- A. Cat 3
- B. Cat 5
- C. Cat 5e
- D. Cat 6A

2. A technician must seat wires into a wall jack's insulation-displacement terminals. Which tool fits?

- A. Punchdown tool
- B. Toner probe
- C. Wi-Fi analyzer
- D. Loopback plug

3. A cable runs through a space used for environmental air. Which property must be checked?

- A. Its color
- B. Its plenum/fire rating
- C. Whether it is Cat 6 only
- D. Whether its connector is blue

Answers - check only after you try

1. D. Cat 6A supports compliant 10GBASE-T channels up to about 328 ft. Cat 6 has more restrictive conditions for 10G.

2. A. A punchdown tool seats and usually trims wires in these terminals. A crimper is used for compatible modular plugs.

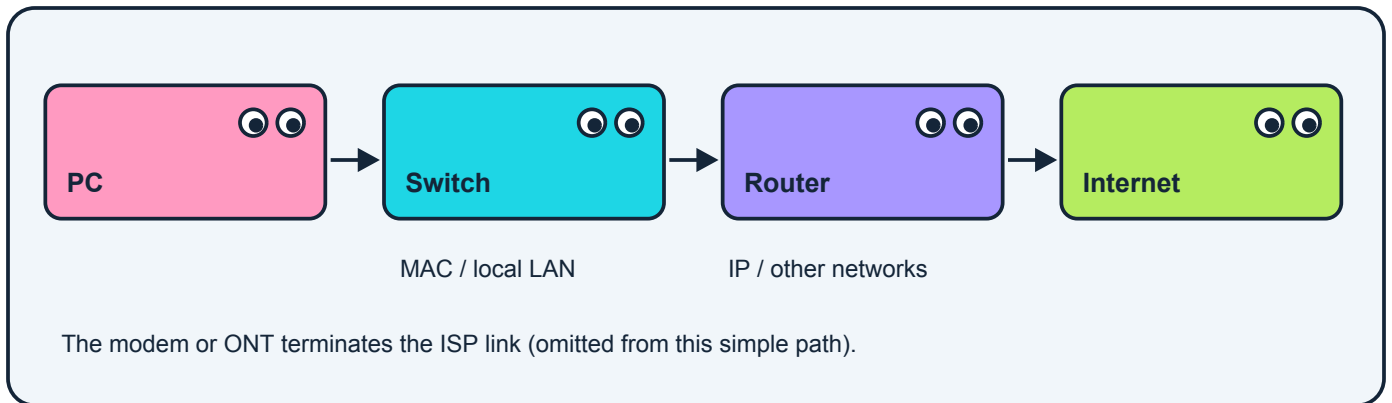
3. B. The applicable fire rating is the clue. Speed category and jacket color do not establish permission for an air-handling space.

Say it back: explain the right answer without naming its letter. Then explain why one other option does not fit. If you can do that, you are learning the concept.

Routers and switches: neighborhood or highway?

Related Core 1 objectives: 2.5, 2.4, 5.5

Visual: A cyan switch mail sorter points to numbered ports. A purple router traffic officer points toward a second network. A green access point sends radio waves to a laptop.



A small office often hides several jobs in one box labeled “wireless router.” Inside may be a router, Ethernet switch, wireless access point, firewall, and DHCP service. Learn the jobs separately so a question about one function does not trick you.

Switch: local delivery. A typical Layer 2 Ethernet switch connects devices within a LAN. LAN means Local Area Network. The switch learns source MAC addresses and their ports, then uses the destination MAC to forward frames. A frame is a link-level data unit.

Router: between networks. A router connects IP networks. It reads the destination IP address and chooses a route. The default gateway is commonly the router interface a host uses to reach other networks. A packet is the IP-level data unit.

Access point: wireless doorway. An AP, or Access Point, bridges wireless clients into a network. It is not automatically a router or modem. A wireless client can join the AP successfully yet still lack valid IP settings or internet service.

Layers as a memory aid. OSI means Open Systems Interconnection. Layer 2 works with local-link addressing such as Ethernet MAC addresses. Layer 3 works with logical addressing such as IP. A multilayer or Layer 3 switch can also route; the word “switch” does not forbid routing features.

WATCH ON YOUTUBE - Professor Messer

[Routers and switches - A+ 220-1201](https://www.youtube.com/watch?v=0hd6_bx0ydo)

https://www.youtube.com/watch?v=0hd6_bx0ydo

The details that separate the answers

Device	Main job	Question clue
Unmanaged switch	Simple plug-and-play local ports.	Few configuration options.
Managed switch	Configurable local networking.	VLANs, monitoring, port settings.
Router	Forward traffic between IP networks.	Different subnets / default gateway.
Access point	Bridge Wi-Fi clients to a LAN.	Add wireless coverage.
Firewall	Permit or deny traffic by policy.	Block unwanted connections.
Patch panel	Organize cable terminations.	Passive wiring; does not switch packets.
Modem / ONT	Terminate the ISP access technology.	Cable/DSL modem or optical network terminal.
PoE switch / injector	Supply power plus Ethernet data.	Power an AP, phone, or camera.

Follow a local packet. PC A and PC B share a subnet and VLAN. PC A learns the neighbor's MAC address using ARP for IPv4, then sends a frame through the switch. A router is not needed for this same-subnet exchange.

Follow a remote packet. PC A compares the destination to its mask and sees a different network. It sends a local frame to its gateway's MAC address. The IP packet still names the remote destination. The router removes the old link frame and forwards using a new link frame. NAT may also change IP information.

VLAN: Virtual Local Area Network. A VLAN separates Layer 2 broadcast domains on shared switch hardware. Two departments can use separate VLANs on one managed switch. Communication between their IP subnets needs routing and suitable policy. A VLAN and a VPN are different: a Virtual Private Network provides a protected logical connection over another network.

PoE: Power over Ethernet. A PoE switch supplies power directly; an injector adds power to a data link. IEEE 802.3af, at, and bt identify PoE families. Common source-power maxima: af 15.4 W, at 30 W, bt Type 3 up to 60 W, Type 4 up to 90 W. Less power reaches the device because of cable loss. Match standards and the total switch power budget.

Troubleshoot in layers. Start with power, cables, link status, and Wi-Fi association. Then inspect IP address, mask, gateway, and DNS. Test local access before remote services. Do not replace the router simply because one browser reports an error.

Can you explain your choice?

Original Core 1-style practice | Select ONE answer per question

EXAM TRAP

A switch floods broadcasts and unknown destinations within the relevant VLAN when appropriate; it does not send every known unicast to every port. A patch panel needs a connection to active network equipment to provide a working network path.

1. Two departments use different IP subnets. Which function lets them communicate?

- A. Punchdown
- B. Routing
- C. Repeating a signal only
- D. File synchronization

2. An office needs separate VLANs and remote port monitoring. Which switch type fits?

- A. Unmanaged
- B. Passive patch panel
- C. Managed
- D. Cable modem

3. An existing wired LAN needs Wi-Fi coverage in a conference room. Which device provides the wireless bridge?

- A. Access point
- B. RAID controller
- C. DNS record
- D. Patch panel alone

Answers - check only after you try

1. B. Traffic between IP subnets needs a routing function. A Layer 3 switch can provide it, but a basic Layer 2 switch alone cannot.

2. C. Managed switches provide configurable features such as VLANs and monitoring. An unmanaged switch focuses on simple connectivity.

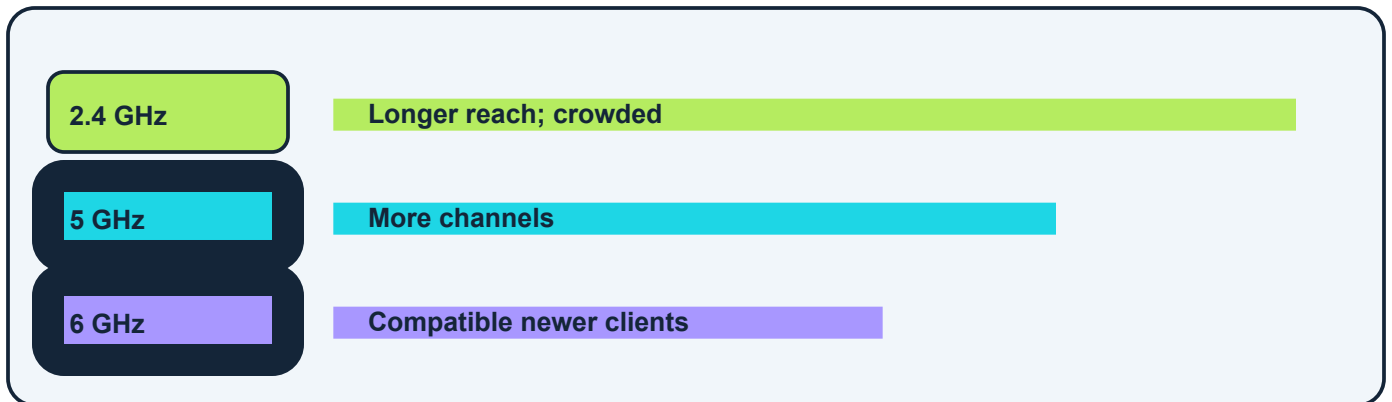
3. A. An AP supplies the wireless connection to the LAN. The existing router can continue to handle traffic between networks.

Say it back: explain the right answer without naming its letter. Then explain why one other option does not fit. If you can do that, you are learning the concept.

Wi-Fi standards: band, channel, speed

Related Core 1 objectives: 2.2, 5.5

Visual: Three neon roads represent 2.4, 5, and 6 GHz. The 2.4 road reaches farther but is crowded. Wider lanes on other bands carry more traffic when conditions allow.



Wi-Fi is wireless local networking based on the IEEE 802.11 family. IEEE means Institute of Electrical and Electronics Engineers. A standard describes supported features. A frequency band is a range of radio frequencies. A channel is a slice of that band. These are related, but they are not the same thing.

2.4 GHz: the longer-reach road. Often reaches farther and passes through obstacles better than higher bands under comparable conditions. It is busy: Wi-Fi, Bluetooth, and other devices can compete or interfere. GHz means gigahertz, a frequency unit, not a data-transfer speed.

5 GHz: more room nearby. Often offers more usable channels and less 2.4 GHz interference. It can support wide channels and fast links, but walls and distance still matter. A device must support the selected band.

6 GHz: newer available space. Wi-Fi 6E adds 6 GHz operation to Wi-Fi 6 capabilities. Both access point and client must support 6 GHz, and local rules control its use. "Wi-Fi 6" alone does not promise 6 GHz support.

Speed is negotiated. An access point's headline rate is not a promise to one laptop. Streams, channel width, signal quality, interference, and client features affect the link. Application throughput is lower than the theoretical radio rate; internet service may be a separate bottleneck.

WATCH ON YOUTUBE - Professor Messer

[Wi-Fi standards - A+ 220-1201](https://www.youtube.com/watch?v=aTuaEk5hnAs)

<https://www.youtube.com/watch?v=aTuaEk5hnAs>

The details that separate the answers

Standard / common name	Band association	Theoretical maximum*
802.11a	5 GHz	54 Mbps
802.11b	2.4 GHz	11 Mbps
802.11g	2.4 GHz	54 Mbps
802.11n / Wi-Fi 4	2.4 or 5 GHz	600 Mbps
802.11ac / Wi-Fi 5	5 GHz	About 6.9 Gbps
802.11ax / Wi-Fi 6	2.4 and 5 GHz	About 9.6 Gbps
Wi-Fi 6E (802.11ax)	Adds 6 GHz capability	Same generation; more spectrum
802.11be / Wi-Fi 7	2.4, 5, and 6 GHz support	Newer context; prioritize band/compatibility clues

Read the speed chart correctly. *These are ideal standard-level maxima, requiring maximum supported configurations; they are not typical laptop speeds. Some devices support only a subset of a standard's bands or features. Learn a/b/g/n/ac/ax and the 6E distinction before chasing marketing numbers.

Channels and widths. In the United States, 1, 6, and 11 are the usual non-overlapping choices for 20 MHz channels in 2.4 GHz. Regulations vary by country. Wider channels can carry more data but consume more spectrum and can be harder to place in a crowded environment.

MIMO and OFDMA. Multiple Input Multiple Output (MIMO) uses multiple radio paths/antennas for spatial streams. Multi-user MIMO serves multiple compatible clients. Orthogonal Frequency Division Multiple Access (OFDMA) divides channel resources among clients efficiently; it is a key Wi-Fi 6 feature.

Troubleshooting clues. Weak signal: inspect distance, obstacles, antenna placement, and coverage. Strong signal but slow service: inspect congestion, interference, uplink and ISP limits. A missing 6 GHz network: verify client hardware, driver, access-point settings, and local availability.

Other radios are different. Bluetooth commonly connects personal devices in a PAN, or Personal Area Network. NFC means Near-Field Communication and supports very short-range exchanges such as tap-to-pay. RFID means Radio-Frequency Identification and uses tags/readers. None is simply another name for Wi-Fi.

Can you explain your choice?

Original Core 1-style practice | Select ONE answer per question

EXAM TRAP

No Wi-Fi standard guarantees one universal range in feet. Walls, antennas, power, local rules, and interference change coverage. Wi-Fi 5 uses 5 GHz; Wi-Fi 6 is a generation name; 5G cellular is a different technology.

1. A laptop supports Wi-Fi 6 only on 2.4 and 5 GHz. Why can it not join a 6 GHz-only network?

- A. It needs 6 GHz-capable hardware
- B. DNS always blocks 6 GHz
- C. Cat 6 cable is required inside the laptop
- D. IPv4 cannot use Wi-Fi

2. Which set is commonly used for non-overlapping 20 MHz channels in U.S. 2.4 GHz Wi-Fi?

- A. 1, 2, 3
- B. 3, 6, 9
- C. 1, 6, 11
- D. 36, 40, 44

3. Which listed standard operates at 5 GHz and is known as Wi-Fi 5?

- A. 802.11b
- B. 802.11g
- C. 802.11n
- D. 802.11ac

Answers - check only after you try

1. A. Wi-Fi 6 does not automatically include 6 GHz. A compatible Wi-Fi 6E or other 6 GHz-capable client is needed, along with software and regional support.

2. C. 1, 6, and 11 is the standard study association in this setting. Nearby channel numbers in 2.4 GHz overlap.

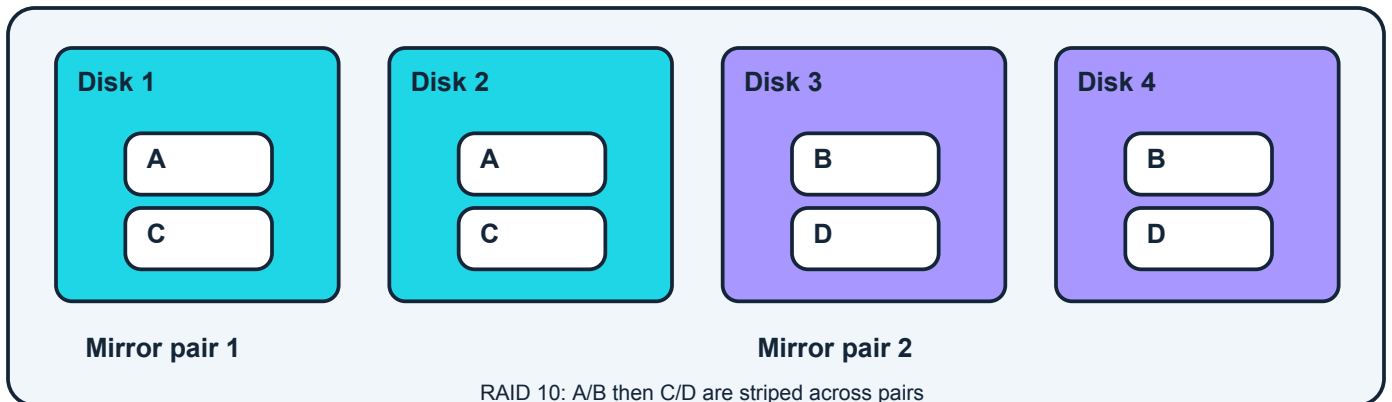
3. D. 802.11ac is Wi-Fi 5 and uses 5 GHz. A dual-band router may offer a separate 2.4 GHz radio using another standard.

Say it back: explain the right answer without naming its letter. Then explain why one other option does not fit. If you can do that, you are learning the concept.

RAID: speed, copies, and recovery math

Related Core 1 objectives: 3.4, 5.2

Visual: Four color-coded drives show RAID 10. Each mirror repeats its teammate's blocks. Follow A, B, C, and D to see how data is striped across the pairs.



RAID means Redundant Array of Independent Disks. An array groups drives into a storage system. Depending on the level, RAID can improve performance, tolerate drive failures, or both. RAID 0 has no redundancy despite the word in the acronym. Learn what the layout does before memorizing its number.

Striping: split the work. Data blocks are spread across drives. Picture splitting a big stack of homework among several desks. RAID 0 stripes data. It can improve performance, but if one drive fails, missing pieces can make the entire array unusable.

Mirroring: keep a matching copy. The same data is written to more than one drive. A basic two-drive RAID 1 mirror has one drive's usable capacity. If one drive fails, the other can still provide the data. Two copies on the same array are not an independent backup.

Parity: recovery information. Parity is calculated information that can help reconstruct missing data. RAID 5 distributes one set of parity across its drives. RAID 6 distributes two independent sets. Parity is not a complete extra copy of every file.

RAID 10: stripe across mirrors. RAID 10, also called 1+0, combines mirrored pairs with striping. It needs at least four drives in the usual layout. It can survive one failed drive in each mirror pair; losing both drives in the same pair loses the array.

WATCH ON YOUTUBE - Professor Messer

[RAID - A+ 220-1201](https://www.youtube.com/watch?v=5E16qftfRY)

<https://www.youtube.com/watch?v=5E16qftfRY>

The details that separate the answers

Level	Minimum drives	Usable capacity*	Drive failures tolerated
0: stripe	2	$N \times S$	None
1: two-drive mirror	2	S	1 of the 2
5: single parity	3	$(N - 1) \times S$	Any 1
6: dual parity	4	$(N - 2) \times S$	Any 2
10: striped mirrors	4; even count	$(N / 2) \times S$	1 per mirror pair; not any 2

Capacity assumptions. *N is the number of active member drives; S is the smallest member's capacity. These common formulas assume matching-size members and the layouts shown. Exclude hot spares. Formatting, metadata, and implementation details reduce reported usable space.

Worked example: four 2 TB drives. TB means terabyte. Raw capacity is 8 TB. RAID 0: 8 TB, no drive protection. RAID 5: $(4 - 1) \times 2 = 6$ TB. RAID 6: $(4 - 2) \times 2 = 4$ TB. RAID 10: $(4 / 2) \times 2 = 4$ TB. Same capacity does not mean the same failure behavior.

RAID 1 example. Two 2 TB drives in a mirror give about 2 TB before overhead, not 4 TB. You spend half the raw space on the matching copy. The files remain available after either one member fails, provided the other member is healthy.

Degraded, spare, rebuild. Degraded means the array has lost some protection or expected operation. A hot spare is a standby replacement; it does not add normal usable capacity. Rebuild recreates missing data on a replacement member. Identify the failed drive precisely and follow the controller's instructions. Hot-swap is allowed only when the system supports it.

RAID versus backup. RAID helps availability after certain hardware failures. It does not undo accidental deletion, ransomware, corruption, theft, or a fire. Those problems can affect all members. Keep separate recoverable backups and test restoration.

RAID 0: split, no copy



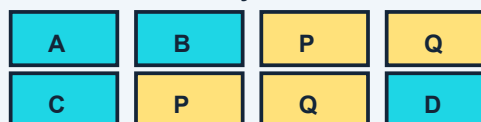
RAID 1: matching copies



RAID 5: P moves across drives



RAID 6: P + Q recovery



Can you explain your choice?

Original Core 1-style practice | Select ONE answer per question

EXAM TRAP

“Survive any two drive failures” points to RAID 6 among these levels. RAID 10 can survive some two-drive combinations, not every combination. Never initialize or delete an existing array as a casual repair step.

1. Four 2 TB drives form RAID 5. What is the usable capacity before overhead?

- A. 2 TB
- B. 4 TB
- C. 6 TB
- D. 8 TB

2. A system must survive the failure of ANY two member drives. Which level fits?

- A. RAID 0
- B. RAID 1 with two drives
- C. RAID 5
- D. RAID 6

3. In RAID 10, both drives in one mirrored pair fail. Another pair remains healthy. What happens?

- A. The array remains fully protected
- B. The array loses data availability
- C. The healthy pair contains every block automatically
- D. A hot spare guarantees recovery

Answers - check only after you try

1. C. One drive-equivalent stores parity: $(4 - 1) \times 2 = 6$ TB. Parity is distributed; it is not necessarily a dedicated physical parity drive.

2. D. RAID 6 has dual parity. A two-drive RAID 1 cannot survive losing both members; RAID 5 tolerates one.

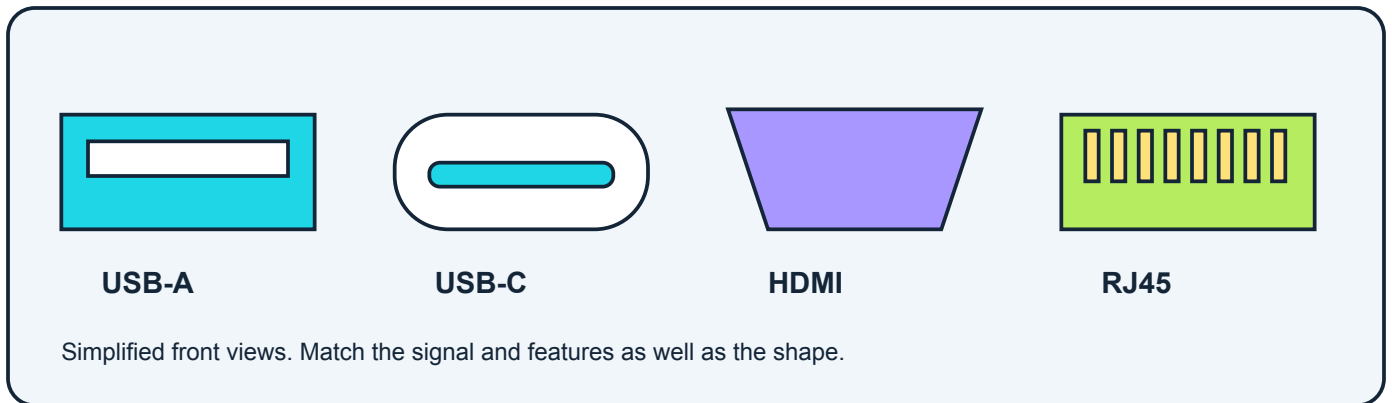
3. B. Striped data was divided between pairs. Losing both copies of one pair removes part of the data. A spare cannot recreate data when the required surviving information is gone.

Say it back: explain the right answer without naming its letter. Then explain why one other option does not fit. If you can do that, you are learning the concept.

Connection types: match shape, signal, and job

Related Core 1 objectives: 3.2, 2.7

Visual: Compare four simplified connector faces: rectangular USB-A, oval USB-C, tapered HDMI, and eight-position RJ45. Shape helps identify the family; it does not prove every feature.



A connector is the physical shape you plug in. An interface or protocol describes how data moves. A version or certification describes supported capabilities. A plug fitting into a socket proves only part of compatibility. Always check the signal, speed, power, direction, and device support. HDMI means High-Definition Multimedia Interface; DVI means Digital Visual Interface; VGA means Video Graphics Array.

USB: Universal Serial Bus. USB connects peripherals and can supply power. Type-A is the familiar rectangular shape. Type-B is often found on printers. Mini-USB and Micro-USB are older small connectors. Type-C is the small reversible oval. Type-C alone does not guarantee a particular speed, display output, or charging wattage.

USB speed names. USB 2.0 is up to 480 Mbps. USB 3.0 is up to 5 Gbps; the same speed is also named USB 3.1 Gen 1 and USB 3.2 Gen 1. USB 3.2 Gen 2 is 10 Gbps; Gen 2x2 is 20 Gbps. Check capability labels instead of assuming all “USB 3” products match.

Thunderbolt. Thunderbolt 1 and 2 used a Mini DisplayPort-shaped connector. Thunderbolt 3 and 4 use USB-C and support up to 40 Gbps link rates. A plain USB-C port is not automatically Thunderbolt. Check the host, cable, and device specifications.

Adapters versus converters. An adapter may change connector shape when compatible signaling exists. An active converter changes signals. A digital-only HDMI output to analog VGA generally needs an active converter; a passive shape-changing cable cannot create an analog signal.

WATCH ON YOUTUBE - Professor Messer

[Connection types - A+ 220-1201](https://www.youtube.com/watch?v=JGuVE-3CvT4)

<https://www.youtube.com/watch?v=JGuVE-3CvT4>

More YouTube lessons: [Video Cables](#) | [Internet Connection Types](#)

The details that separate the answers

Connector / interface	Recognize it	Common purpose
RJ45 (8P8C)	8-position modular plug with latch.	Twisted-pair Ethernet.
RJ11	Smaller telephone-style modular plug.	Telephone / DSL line.
F-type	Round threaded coax connector.	Cable broadband / television.
LC / SC / ST	Small latch / square push-pull / twist-lock.	Fiber patch connections.
HDMI	Flat, tapered digital connector.	Digital video and audio.
DisplayPort	Full size has one beveled corner.	Digital video and audio.
DVI	Larger connector with pin grid.	DVI-D digital; DVI-I also supports analog.
VGA / DE-15	15 pins in three rows.	Analog video, no native audio.
DB9 / DE-9	9 pins in two rows.	Common legacy serial connection.
SATA	Keyed L-shaped connectors.	7-pin data; separate 15-pin drive power.

Storage and power connectors. SATA means Serial Advanced Technology Attachment. SATA revision 3 links support 6 Gbps signaling, not 6 GB/s file transfer. eSATA is external SATA; ordinary eSATA does not inherently supply power. A legacy 4-pin peripheral Molex connector supplies power, not data. Lightning is an Apple connector used by many older mobile devices.

Fiber names. For A+ identification, LC is commonly expanded as Lucent Connector, SC as Subscriber Connector, and ST as Straight Tip. Other industry expansions exist. Recognize their shape and matching fiber/transceiver requirements; the acronym alone does not determine speed.

Internet connection choices. Cable uses coax to a modem. DSL, Digital Subscriber Line, uses telephone copper and is affected by line conditions and distance. Fiber uses optical links and an ONT, Optical Network Terminal. Cellular uses a mobile carrier; tethering shares that connection with another device.

Wireless ISP and satellite. A Wireless Internet Service Provider (WISP) commonly delivers fixed wireless service to a building. Satellite can serve remote places without local wired service. Geostationary satellite has high propagation delay; low-Earth-orbit service can have lower delay. Do not treat every satellite system as identical.

Network size names. LAN is a local network; WLAN is a wireless LAN. WAN is a Wide Area Network linking distant areas. MAN is a Metropolitan Area Network. PAN is a short personal-device network. SAN is a Storage Area Network providing storage connectivity, commonly at block level; a NAS device shares files over a network.

Can you explain your choice?

Original Core 1-style practice | Select ONE answer per question

EXAM TRAP

Video over USB-C requires a supported mode such as DisplayPort Alt Mode or suitable Thunderbolt/USB capabilities. A charging-only cable cannot carry a monitor image. Color alone is not a reliable connector specification.

1. A USB-C charging cable powers a laptop but a monitor stays blank. What must be checked?

- A. Only the cable color
- B. Whether the path supports video
- C. Whether IPv6 is enabled
- D. The printer queue

2. Which connector usually attaches the coax line to a cable modem?

- A. RJ11
- B. LC
- C. F-type
- D. USB-B

3. A digital-only HDMI source must drive an analog VGA display. What is generally required?

- A. An active converter
- B. A passive gender changer only
- C. A DHCP reservation
- D. A SATA power splitter

Answers - check only after you try

1. B. Verify video support in the laptop port, cable, adapter, and monitor. The Type-C shape and charging success do not prove video capability.

2. C. F-type is a threaded coax connector. RJ11 is commonly telephone/DSL; LC is fiber.

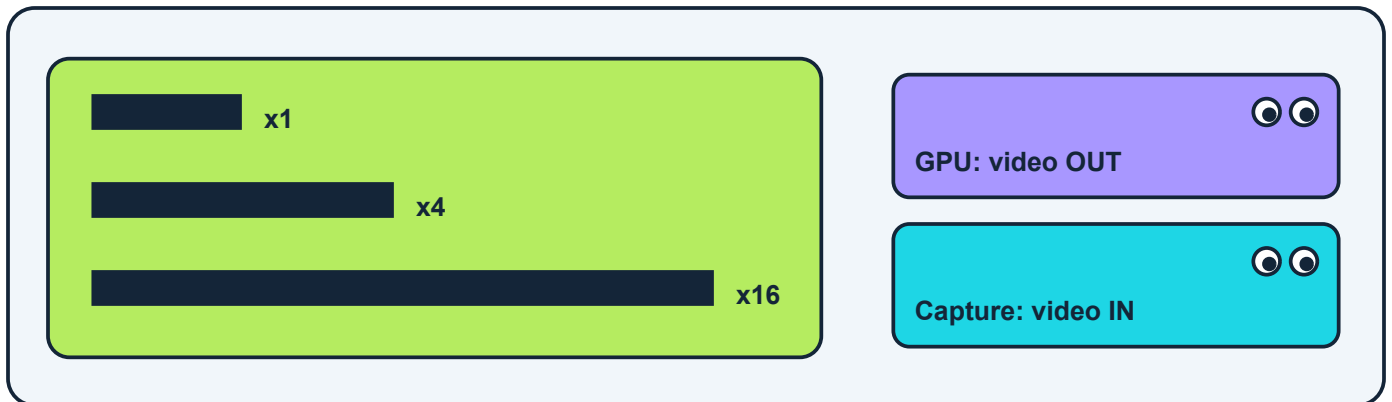
3. A. The signal must change from digital to analog. Also verify direction and power requirements; some converters work only one way.

Say it back: explain the right answer without naming its letter. Then explain why one other option does not fit. If you can do that, you are learning the concept.

PCI cards: give your computer a new ability

Related Core 1 objectives: 3.5, 3.4

Visual: A green motherboard shows short and long PCIe slots. The x labels identify lane widths. Two adapter characters remind you: a GPU sends video out; a capture card brings video in.



An expansion card adds a function to a computer. It plugs into a compatible motherboard connector and exposes ports or hardware features. PCI means Peripheral Component Interconnect. PCI Express, written PCIe, is the modern serial expansion interface. Conventional PCI and PCIe are not interchangeable slots.

A lane is a data path. PCIe uses point-to-point links made of lanes. Common widths are x1, x4, x8, and x16. “x4” means four lanes, not version 4. Each lane supports communication in both directions. More lanes can provide more total bandwidth.

Generation is separate. PCIe 3.0, 4.0, and 5.0 are generations. Newer generations increase per-lane speed. A PCIe 4.0 x4 device uses generation 4 signaling over four lanes. It is not the same description as a PCIe 3.0 x16 device.

Physical size versus electrical lanes. A long x16-shaped slot may be wired for fewer lanes. A smaller PCIe card generally works in a compatible larger PCIe slot, but check the motherboard manual. A large card will not fit every small closed slot. Lane sharing can reduce bandwidth or disable another connector.

Backward compatibility. PCIe generations are designed to interoperate, usually at the highest generation supported by both ends. Physical fit, firmware support, operating-system drivers, power, and lane configuration still matter. Do not force a card into a slot.

WATCH ON YOUTUBE - Professor Messer

[PCI cards - A+ 220-1201](https://www.youtube.com/watch?v=KtipbKR8rqo)

<https://www.youtube.com/watch?v=KtipbKR8rqo>

The details that separate the answers

Card or component	Purpose	Scenario clue
NIC: Network Interface Card	Add Ethernet or Wi-Fi connectivity.	Missing/failed network port; faster network required.
GPU: Graphics Processing Unit	Render graphics; drive display outputs.	Graphics workload; connect monitor to active card output.
Capture card	Bring external video into the computer.	Record a camera or game console.
Sound card	Audio input/output features.	Special audio connections or capabilities.
USB expansion card	Add supported USB controllers/ports.	Need additional peripheral connections.
Storage controller / HBA	Connect and manage supported drives.	Additional storage interfaces; HBA = Host Bus Adapter.
M.2 module	Compact form factor, not a speed promise.	May carry SATA or PCIe/NVMe, depending on module and slot.

M.2 versus NVMe. M.2 describes a module/connector form factor. NVMe means Non-Volatile Memory Express, a storage protocol commonly carried over PCIe. An M.2 drive can instead use SATA. Match keying, supported protocol, and length. An M.2 2280 module is about 0.87 in wide by 3.15 in long; 2280 remains its standard size code.

Plan before installing. Read the motherboard and card requirements. Check slot type, electrical lanes, physical clearance, case bracket height, power supply capacity, auxiliary power connectors, and operating-system support. Large GPUs may cover neighboring slots.

Install safely. Shut down, disconnect external power, and follow the service manual for battery and residual-power steps. Use appropriate electrostatic discharge (ESD) protection. Hold a card by its edges, align the notch, seat it evenly, and secure its bracket. Do not service inside a power supply.

Driver and verification. A driver lets the operating system communicate with hardware. Follow the maker's order for software installation. After installation, verify detection in firmware/Device Manager, install a suitable trusted driver, connect required external antennas or cables, and test the intended function.

Troubleshooting clues. No detection: check seating, supported slot, firmware settings, power, and lane-sharing rules. Detected with an error: inspect driver and resources. New GPU but blank screen: check the selected monitor input and which output is connected, then seating and power. Disconnect power before reseating.

Can you explain your choice?

Original Core 1-style practice | Select ONE answer per question

EXAM TRAP

A graphics card outputs a picture; a capture card receives video. PCIe x16 is a lane-width description, not a guarantee of generation, actual wiring, or a particular number of frames per second.

1. A computer must record video from an external camera's HDMI output. Which card fits?

- A. Sound card
- B. Capture card
- C. Ethernet NIC
- D. Basic display-output adapter

2. What does x4 mean in "PCIe 4.0 x4"?

- A. Four USB ports
- B. Fourth-generation CPU
- C. Four PCIe lanes
- D. Four inches long

3. An M.2 SATA SSD fits a slot that supports only PCIe/NVMe storage. Is physical fit enough?

- A. Yes, all M.2 drives use NVMe
- B. Yes, a driver changes SATA to NVMe
- C. Only if it is painted green
- D. No; protocol support must match

Answers - check only after you try

1. B. The computer needs video input, which is the capture card's job. A normal display output does not become an input just because both use HDMI.

2. C. 4.0 is the generation; x4 is the lane count. Keeping those two labels separate prevents a common compatibility mistake.

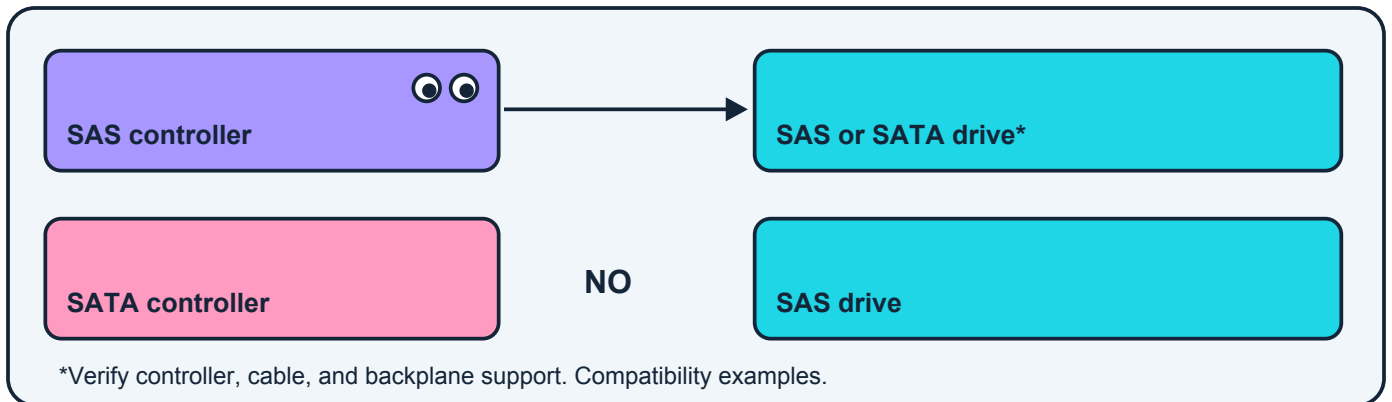
3. D. M.2 is a form factor. The host must support the drive's actual interface/protocol; physical fit alone is not enough.

Say it back: explain the right answer without naming its letter. Then explain why one other option does not fit. If you can do that, you are learning the concept.

SCSI and SAS: speak the drive's language

Related Core 1 objectives: 3.4, 3.5; legacy context labeled

Visual: A controller sends commands to drives. A serial SAS path and a SATA path look similar, but the controller must speak the right language.



SCSI means Small Computer System Interface, pronounced “scuzzy.” It is a family of standards and commands for communicating with devices, especially storage. A command says what to do, such as read a block. The connection transports that command. SCSI does not name one universal cable, speed, or drive shape.

Parallel SCSI: the older shared road. Older SCSI systems used a shared parallel bus, often with several devices on a cable. Devices needed unique SCSI IDs, and the bus needed proper termination at its two ends. This is legacy context: do not apply its jumper and termination rules to a modern SAS drive.

SAS: Serial Attached SCSI. SAS carries SCSI commands over serial links. It is common in servers, storage arrays, and enterprise drives. A SAS controller or Host Bus Adapter (HBA) connects the system to supported devices. SAS can use direct connections, backplanes, and expanders to reach more drives.

SAS versus SATA compatibility. A compatible SAS controller can often use SATA drives if the controller, cable, backplane, and software support them. A SATA-only controller cannot run a SAS drive. A passive connector adapter cannot teach a SATA controller the SAS protocol.

Compare SATA and NVMe. SATA, Serial Advanced Technology Attachment, is common for client drives and some server storage. NVMe, Non-Volatile Memory Express, is a different storage command protocol designed for nonvolatile memory, commonly using PCIe. An M.2 shape does not prove NVMe or SATA support.

WATCH ON YOUTUBE - Professor Messer

[Storage Devices - A+ 220-1201](https://www.youtube.com/watch?v=UfLNi8-c9VM)

<https://www.youtube.com/watch?v=UfLNi8-c9VM>

The details that separate the answers

Term	What it describes	Do not confuse it with...
SCSI	Command/interface standards family.	One fixed transfer speed.
Parallel SCSI	Legacy shared parallel connection.	Modern point-to-point SAS wiring.
SAS	Serial Attached SCSI transport/interface.	SATA-only controller support.
SATA	Serial ATA storage interface.	The speed of every attached drive.
NVMe	Storage protocol commonly over PCIe.	M.2, which describes a form factor.
HBA	Host Bus Adapter: connects host to devices.	Guaranteed RAID capability.
RAID controller	Controller that can implement RAID.	A backup system.
iSCSI (extra context)	SCSI commands transported over TCP/IP.	A USB connector or ordinary file share.

Speed depends on the generation. Legacy Ultra320 parallel SCSI has a nominal shared-bus rate of 320 MB/s. SAS examples include 6 Gb/s and 12 Gb/s per link; newer generations exist. SATA revision 3 uses 6 Gb/s signaling. Never interpret “SCSI” alone as a specific maximum speed.

Legacy IDs and termination. On classic narrow parallel SCSI, IDs commonly range from 0 to 7; wide SCSI commonly uses 0 to 15. The host adapter also needs an ID. Termination belongs at the bus ends, not at every drive. Follow the exact legacy equipment manual; connector and signaling variants differ.

SAS installation. Modern SAS does not require the old parallel-SCSI ID jumpers or manual end terminators on each drive. Check the controller’s supported drive types, ports, cables/backplane, power, firmware, and driver. A drive behind a controller may appear in that controller’s management utility rather than the motherboard’s SATA list.

SAS is not automatically faster in every task. The interface provides a possible data path. A rotating SAS hard drive can still be slower than an SSD with a different interface for a given workload. Compare actual device performance, latency, and workload, not the label alone.

How an A+ scenario points you. “Enterprise serial SCSI drive” points to SAS. “Will this SAS drive work on a SATA-only motherboard?” points to incompatible controller support. “Module fits but is not detected” points to checking the form factor AND protocol. Legacy termination details are background, not the main Core 1 focus.

Can you explain your choice?

Original Core 1-style practice | Select ONE answer per question

EXAM TRAP

A SAS connector fitting a cable is not enough: the host must support SAS. Support is directional: many SAS controllers accept SATA, but SATA-only controllers do not accept SAS. Always verify the specific equipment.

1. A technician buys a SAS drive for a desktop with SATA-only storage ports. What is needed to use it?

- A. A passive shape adapter only
- B. A compatible SAS controller and supported connection path
- C. A new DNS record
- D. A SATA driver update that converts it to SAS

2. A server uses a supported SAS controller and backplane. Which statement about a SATA drive is accurate?

- A. It may work if the controller and path support it
- B. SATA can never work behind SAS
- C. Every SATA drive has dual SAS ports
- D. It becomes an NVMe drive

3. Which item is a storage protocol rather than a module's physical shape?

- A. M.2
- B. 2.5-inch
- C. 3.5-inch
- D. NVMe

Answers - check only after you try

1. B. The host must speak SAS. A compatible controller plus suitable cabling, backplane, power, and software provides that support. Changing connector shape alone does not.

2. A. SAS controllers commonly support SATA devices, but you must verify the specific system. This does not change the drive into SAS or NVMe.

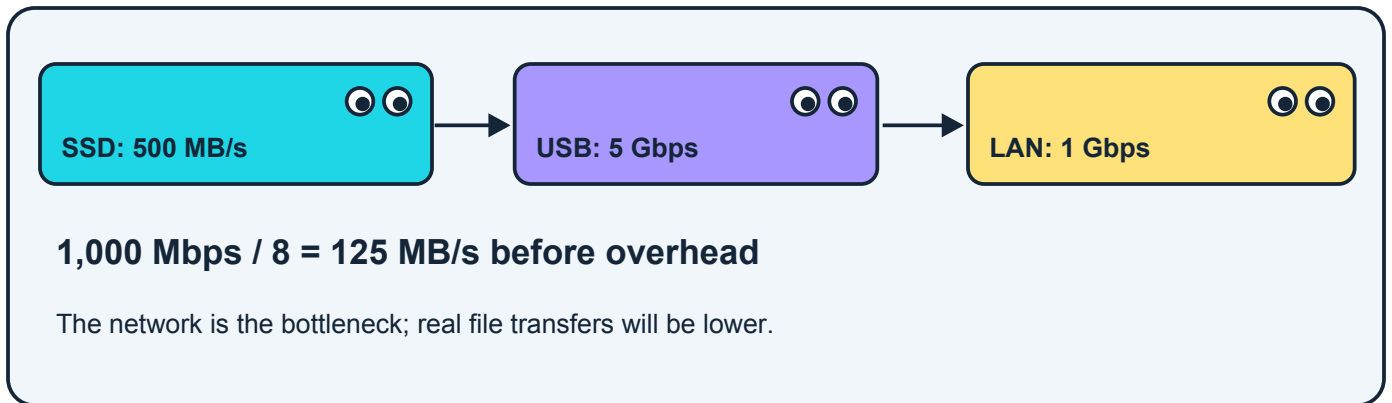
3. D. NVMe specifies how storage commands are handled. M.2 and inch-size descriptions identify form factors, which do not alone prove protocol compatibility.

Say it back: explain the right answer without naming its letter. Then explain why one other option does not fit. If you can do that, you are learning the concept.

Connection throughput: find the slowest link

Related Core 1 objectives: 3.2, 3.4, 3.5, 5.5

Visual: A fast SSD pours data into a wide USB pipe, but a narrow 1 Gbps network pipe limits the final flow. Eight bit tokens fill one byte box.



Throughput is how much data actually moves in a period of time. Bandwidth is the path's capacity. Latency is the delay before or during a response. A wide highway can move many cars, but a long trip still takes time. A fast connection label does not promise the same file-copy speed.

Lowercase b versus uppercase B. b means bit; B means byte. One byte contains 8 bits. Mbps means millions of bits per second; MB/s means millions of bytes per second. Gbps and GB/s use billions. The capital letter changes the answer by a factor of eight.

Convert the units first. To convert bits per second to bytes per second, divide by 8. A nominal 1 Gbps link is $1,000 \text{ Mbps} / 8 = 125 \text{ MB/s}$ before overhead. A measured 80 MB/s transfer is 640 Mbps. These examples use decimal units; MiB and GiB are different binary units.

The bottleneck rule. The slowest relevant part limits the path. A fast SSD through a 5 Gbps USB adapter and then a 1 Gbps network link cannot deliver its full SSD rate across that network. Protocol overhead, contention, and the destination drive can reduce the result further.

Throughput versus IOPS. Input/Output Operations Per Second (IOPS) counts operations, not bytes. Large sequential transfers can show high MB/s. Many tiny random reads depend strongly on latency and IOPS. A high link rate cannot remove a hard drive's mechanical seek delay.

WATCH ON YOUTUBE - Professor Messer

[Peripheral Cables - A+ 220-1201](https://www.youtube.com/watch?v=JGuVE-3CvT4)

<https://www.youtube.com/watch?v=JGuVE-3CvT4>

The details that separate the answers

Connection / version	Nominal link or bus rate	Read the number correctly
USB 2.0	480 Mbps	60 MB/s arithmetic ceiling; actual lower.
USB 3.0 / 3.2 Gen 1	5 Gbps	Same 5 Gbps speed under different names.
USB 3.2 Gen 2 / Gen 2x2	10 / 20 Gbps	Both ends and cable must support the mode.
Thunderbolt 1 / 2 / 3-4	10 / 20 / 40 Gbps	Generation matters; shape alone is not proof.
SATA revisions 1 / 2 / 3	1.5 / 3 / 6 Gbps	SATA 6 Gb/s is not 6 GB/s.
SAS examples	6 / 12 Gbps per link	Examples, not the latest possible SAS ceiling.
Ultra320 parallel SCSI	320 MB/s shared bus	Already bytes; do not divide by 8 again.
Ethernet examples	100 Mbps / 1 Gbps / 10 Gbps	12.5 / 125 / 1,250 MB/s before overhead.
PCIe 3.0 x4 / 4.0 x4	About 3.94 / 7.88 GB/s each direction*	After encoding, before packet overhead.

Why the arithmetic ceiling is not a promise. Signaling can include encoding bits, packet headers, acknowledgements, and other work. SATA 6 Gb/s uses 8b/10b encoding, leaving about 600 MB/s before additional overhead. Many SATA SSDs achieve roughly 500-550 MB/s sequentially; a spinning drive may be much slower.

Worked bottleneck example. Assume source storage can read 500 MB/s and destination can write 300 MB/s. The network is 1 Gbps, or 125 MB/s before overhead. Minimum of 500, 300, and 125 is 125 MB/s. Actual file transfer will be below that network ceiling.

Worked time example. A 2,000 MB file transfers at a measured steady 100 MB/s. Time = size / rate = 2,000 / 100 = 20 seconds. Use matching units. If the question gives only a nominal link rate, your calculated time is an ideal minimum, not guaranteed elapsed time.

PCIe generations and lanes. *PCIe 3.0 sends 8 GT/s per lane; 4.0 sends 16 GT/s. GT/s means billions of transfers per second. Both use 128b/130b encoding. Four lanes give about 3.94 or 7.88 GB/s per direction before packet overhead. This is interface capacity, not an SSD benchmark.

Check the complete path. Check the host port, cable, hub, adapter, drive, destination, and shared uplink. Devices negotiate a common supported mode. HDMI/DisplayPort bandwidth also depends on version, resolution, refresh rate, and supported modes.

Can you explain your choice?

Original Core 1-style practice | Select ONE answer per question

EXAM TRAP

“USB-C” is a connector shape, not a throughput number. “6 Gb/s SATA” is not 6 GB/s, and simply dividing by eight still ignores encoding. Compare all rates in the same units before selecting the bottleneck.

1. Ignoring all overhead, what byte rate corresponds to 1 Gbps?

- A. 1 MB/s
- B. 8 MB/s
- C. 125 MB/s
- D. 1,000 MB/s

2. Storage can supply 500 MB/s, but the path includes a 100 Mbps network link. What is the ideal link ceiling?

- A. 12.5 MB/s
- B. 100 MB/s
- C. 500 MB/s
- D. 800 MB/s

3. A 3,000 MB file transfers at a measured 150 MB/s. About how long does it take?

- A. 2 seconds
- B. 20 seconds
- C. 150 seconds
- D. 450 seconds

Answers - check only after you try

- 1. C.** 1 Gbps is 1,000 Mbps. Divide by eight bits per byte: 125 MB/s. Actual application throughput is lower.
- 2. A.** 100 Mbps / 8 = 12.5 MB/s. The network is the bottleneck; the source drive’s higher rate cannot bypass it.
- 3. B.** 3,000 / 150 = 20 seconds. Because a measured throughput was supplied, no bit-to-byte conversion is needed here.

Say it back: explain the right answer without naming its letter. Then explain why one other option does not fit. If you can do that, you are learning the concept.

Your fast comparison sheet

When the question says...	Think...
Rent a VM and manage its operating system	IaaS
Deploy code without managing its OS/runtime	PaaS
Use a complete provider-managed app	SaaS
One organization / shared special community / joined clouds	Private / community / hybrid
Resources rise AND fall with demand	Elasticity
Lease an address / resolve a name	DHCP / DNS
Move within a LAN / between IP networks	Switch / router
Add Wi-Fi to a wired LAN	Access point
169.254 after automatic IPv4 setup	Investigate link and DHCP reachability
172 private range	172.16 through 172.31
10G copper near the full 328 ft channel	Cat 6A
Wi-Fi 5 / Wi-Fi 6 / Wi-Fi 6E	ac at 5 GHz / ax / ax with 6 GHz capability
Any two drive failures	RAID 6
Striping / mirroring / dual parity	RAID 0 / RAID 1 / RAID 6
4 x 2 TB in RAID 5 / RAID 10	6 TB / 4 TB before overhead
USB-C fits, but video does not work	Check video mode, cable, adapter, and host
Record external video / render display output	Capture card / GPU
PCIe 4.0 x4	Generation 4; four lanes
M.2 fits but drive is not detected	Check SATA versus PCIe/NVMe support

READ THE QUALIFIER

Underline the requirement: FIRST, BEST, MOST likely, minimum, maximum, or Select TWO. Choose what meets the stated facts. A possible answer can still be weaker than the one that matches all the requirements.

Primary sources and lesson links

The official objectives set the scope. The references below support technical explanations. Video links were matched to Professor Messer's 220-1201 lesson pages. The illustrations are original teaching diagrams; connector drawings are simplified, not engineering drawings.

[Official CompTIA A+ 220-1201 objectives, version 3.0](#)

1. Cloud types

[NIST: Definition of cloud computing](#)

[Microsoft: Shared cloud responsibility](#)

[Professor Messer: matching video lesson and transcript](#)

2. Server types

[Microsoft: DNS overview](#)

[Microsoft: DHCP overview](#)

[Microsoft: DNS architecture and resource records](#)

[IANA: Service names and port numbers](#)

[IETF: Syslog, RFC 5424](#)

[Cisco: Authentication, authorization and accounting](#)

[Professor Messer: matching video lesson and transcript](#)

3. IP addresses

[IETF: Private IPv4 address ranges, RFC 1918](#)

[IETF: IPv4 link-local addresses, RFC 3927](#)

[Microsoft: TCP/IP addressing and subnetting basics](#)

[Microsoft: ipconfig command](#)

[Professor Messer: matching video lesson and transcript](#)

Primary sources and lesson links

4. Cable categories

[Cisco: Catalyst 1000 hardware and PoE specifications](#)

[Fluke Networks: 10GBASE-T field testing requirements](#)

[Belden: Communications cable ratings](#)

[FOA: Fiber optic connector types](#)

[Klein Tools: Telecom tools catalog](#)

[Professor Messer: matching video lesson and transcript](#)

5. Routers and switches

[Cisco: Networking basics](#)

[Cisco: VLAN overview](#)

[Cisco: Catalyst 1000 hardware and PoE specifications](#)

[Verizon: Optical Network Terminal](#)

[Cisco: IEEE PoE types and power budgets](#)

[Professor Messer: matching video lesson and transcript](#)

6. Wi-Fi standards

[Intel: Wi-Fi protocols and data rates](#)

[Cisco: Wi-Fi 6E band operation and client connectivity](#)

[Cisco: Aironet 1850 radio specifications](#)

[Cisco: Wireless RF reference guide](#)

[Cisco: Wi-Fi generation capabilities and maximum rates](#)

[Professor Messer: matching video lesson and transcript](#)

Primary sources and lesson links

7. RAID

[Dell: RAID levels and specifications](#)

[Intel: Defining RAID volumes](#)

[Dell: PowerEdge RAID controller H740P manual](#)

[Professor Messer: matching video lesson and transcript](#)

8. Connection types

[USB-IF: USB 3.2 transfer rates](#)

[USB-IF: USB Type-C language guidelines](#)

[Intel: Thunderbolt technology overview](#)

[Dell: External ports and connectors](#)

[Microsoft: Free up drive space in Windows](#)

[FOA: Fiber optic connector types](#)

[Verizon: DSL self-installation and filters](#)

[ESA: Satellite orbits](#)

[Professor Messer: matching video lesson and transcript](#)

9. PCI cards

[Intel: PCIe generation and lane explanation](#)

[Kingston: SATA and M.2 SSD FAQ](#)

[Microsoft: Update drivers through Device Manager](#)

[Intel: CPU and GPU functions](#)

[Dell: Service safety and standby power](#)

[Professor Messer: matching video lesson and transcript](#)

Primary sources and lesson links

10. SCSI and SAS

[Seagate: SAS frequently asked questions](#)

[Seagate: SAS installation and automatic device identification](#)

[Seagate: SATA drive and SAS controller compatibility](#)

[Seagate: Cheetah 10K.6 legacy Ultra320 SCSI manual](#)

[Professor Messer: matching video lesson and transcript](#)

11. Connection throughput

[Seagate: Interface bandwidth versus actual transfer rates](#)

[USB-IF: USB 3.2 transfer rates](#)

[Intel: PCIe generation and lane explanation](#)

[Kingston: SATA and M.2 SSD interface limits](#)

[Professor Messer: matching video lesson and transcript](#)